328654(28)

B. E. (Sixth Semester) Examination Nov.-Dec. 2019

(New Scheme)

(Et & T Branch)

VLSI DESIGN

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) in each question is compulsory which is of 2 marks. Attempt any* ***two*** *parts from the rest three, each is of 7 marks.*

Unit - I

1. (a) Define propagation delay time of an inverter circuit.

   (b) Find the delay for the case where the internal FET capacitance is C_{FET} = 80 fF. The rise time tr is

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controlled by the PFET that has a resistance of $R_p = 822 \cdot 9\ \Omega$.

(c) Estimate the two input NAND circuit rise and fall time in terms of resistances and capacitances.

(d) Draw Transmission gate model and explain it's working in terms of delay by internal resistances and capacitances.

Unit - II

2. (a) What are the three types of silicon wafers?

(b) Explain in detail material deposition and lithography used in fabrication process of integrated circuit.

(c) Draw stick and layout diagram of three input NAND circuits.

(d) Explain with suitable layout, active contacts and metal vias.

Unit - III

3. (a) What is the role of capacitance C_m in 1T DRAM?

(b) What is parity generator? Draw its basic block diagram and stick diagram.

(c) Draw stick diagram and CMOS layout for the given AOI expression

$$F = \overline{x \cdot y + z \cdot w}$$

(d) Draw exclusive – OR mirror circuit with its CMOS layout.

Unit - IV

4. (a) What do you mean by "MOD" and "REM" operators?

(b) Write a VHDL code for BCD to 7 segment decoder and 2 to 4 binary decoder using "Case statement".

(c) Explain following "WAIT" statements with suitable examples :

(i) WAIT ON

(ii) WAIT UNTIL

(iii) WAIT FOR

What is the difference between statements :

(i) FOR LOOP

(ii) FOR Generate

Explain with example.

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(d) Distinguish the following :

(i) Concurrent and sequential process

(ii) Sensitive and desensitive process

(iii) Operators and overloading operators

Unit - V

5. (a) What is the difference between Moore and Mealy finite state machine?

(b) Write a VHDL code for an n-bit left to right shift register using for LOOP.

(c) Write VHDL code for a sequence detector "10011" using Moore FSM.

(d) Design full adder using half adder and write a test bench.

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